

# Ethan Stace

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## Education

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| <b>University of Florida</b><br><i>BS in Physics, Mathematics, Minor in Electrical Engineering (GPA: 3.93/4.00)</i> <ul style="list-style-type: none"><li>• <b>Relevant Coursework:</b> Solid State Physics, Solid State Devices, Statistical Mechanics, Quantum Mechanics I &amp; II, Circuits, Photonics, MEMS Transducers, Electromagnetism I &amp; II, Spintronics, Machine Learning for Physics</li></ul> | August 2023 - May 2027<br>Gainesville, FL |
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## Experience

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| <b>Undergraduate Researcher in Electrical Engineering</b><br><i>University of Florida</i> <ul style="list-style-type: none"><li>• Advisor: Prof. Philip Feng (Electrical &amp; Computer Engineering)</li><li>• Developing 4-node Ising machine using NEMS resonators to solve Max-Cut problems</li><li>• Characterized and tested NEMS resonators using probe station, lock-in amplifier, and oscilloscope</li><li>• Designing PCB to miniaturize Ising machine on KiCad software</li><li>• Fabricating thin-film resonators using cleanroom techniques for low-temperature measurements in Microkelvin Laboratory</li></ul>                                                 | November 2025 - Present<br>Gainesville, FL |
| <b>Penn State 2DCC/Nanofab REU Intern</b><br><i>The Pennsylvania State University</i> <ul style="list-style-type: none"><li>• Advisors: Prof. Saptarshi Das, Prof. Suzanne Mohney (Electrical &amp; Computer Engineering)</li><li>• Assisted in fabrication and characterization of 2D field-effect transistors using cleanroom and lab equipment. Shadowed electron-beam lithography, electron-beam deposition, atomic layer deposition, and reactive ion plasma etching</li><li>• Performed rapid thermal annealment and characterization of ohmic contacts to semiconductor thin films</li><li>• Analyzed XRD data of ohmic contacts to AlN using Jade software</li></ul> | May 2026 - July 2026<br>State College, PA  |
| <b>UT-Dallas Physics REU Intern</b><br><i>The University of Texas at Dallas</i> <ul style="list-style-type: none"><li>• Advisor: Prof. Aaron Smith (Astrophysics)</li><li>• Developed Atom module in Cosmic Lyman-<math>\alpha</math> Transfer Code, a new module for simulating multi-lined, coupled resonant scattering in astrophysical media</li><li>• Developed novel analytical solution to radiative transfer equation to model multi-lined transfer networks</li><li>• Used Atom module to study Lyman-<math>\alpha</math> and Deuterium-injected systems</li></ul>                                                                                                  | May 2025 - August 2025<br>Dallas, TX       |
| <b>Undergraduate Researcher in Plasma Astrophysics</b><br><i>University of Florida</i> <ul style="list-style-type: none"><li>• Advisor: Prof. Siyao Xu (Physics)</li><li>• Analyzed data from HPC simulations to study physics of weakly-collisional positron-electron plasma</li><li>• Developed novel algorithm to measure mean free path by tracking sharp changes in trajectories, and how the mean free path changed at different length and energy regimes</li><li>• Showed that mean free path distribution is non-Maxwellian, matching expected behavior from anisotropy of weakly-collisional plasmas.</li></ul>                                                    | October 2024 - May 2025<br>Gainesville, FL |

## Extracurricular Activities

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| <b>Society of Physics Students   Outreach Chair, (3x)</b> <ul style="list-style-type: none"><li>• Lead the outreach efforts of the Society of Physics Students, both within the department and with outside organizations</li><li>• Created the outreach committee to brainstorm ideas and organize events</li><li>• Lead the Mentor-Mentee program, responsible for Mentor-Mentee speed-dating, setting up social events, and ensuring healthy communication between matches</li></ul> | Fall 2024 - May 2027 |
| <b>SCEECs Summer School on Extreme Electrodynamics and Plasma Physics   Student</b>                                                                                                                                                                                                                                                                                                                                                                                                     | June 2025            |

- Attended intensive two weeks of graduate-level lectures on plasma astrophysics at The University of Texas at San Antonio taught by renowned scientists from academic institutions across the world

### **SIMIODE Challenge Using Differential Equations Modeling | *Competitor***

November 2023

- Joined a three-person team to compete in an international math-modeling competition
- Created infectious disease model using Lotka-Volterra equations to model the spread of violence through human populations, then analyzed the behavior of solutions using numerical methods
- Scored the highest raw score (4.857/5.000) out of all 124 teams, earned *Outstanding Team Award*

## **Publications, Posters, and Presentations**

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### **Publications**

- Caplice, T., Kaiser, T., **Stace, E.**, Shaw, S., Casella, C., Feng, P.X.-L. "A Nanoelectromechanical Oscillator Ising Machine for Solving Max-Cut Problems." *Nature Microsystems and Nanoengineering*. (Submitted)
- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman- $\alpha$  Fine Structure and Deuterium Coupling." *Publications of the Astronomical Society of Australia*. <https://arxiv.org/abs/2511.20580>. (Accepted 1/22/26)

### **Conference Papers**

- Caplice, T., Kaiser, T., **Stace, E.**, Roukes, M., Casella, C., Shaw, S., Feng, X. L. "Experimental Demonstration of Nanomechanical Oscillator Ising Machine for Solving Optimization Problems." *Hilton Head Workshop 2026*.

### **Presentations**

- Randazzo, A., Prince, N., **Stace, E.**, Burrowes, A. "Atomistic Spin Simulations with Sunny.jl." *Introduction to Solid State Physics, 2026*
- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman- $\alpha$  Fine Structure and Deuterium Coupling." *2026 APS Global Physics Summit*.
- **Stace, E.** "A Short Review of the Feynman Path Integral." *Introduction to Particle Physics, 2025*
- **Stace, E.** "Predicting Wheat Diseases using CNNs." *Machine Learning for Physics, 2024*

### **Posters**

- **Stace, E.**, Alam, M.D.S., Nam, T., Das, S. "Threshold Engineering for 2D Field-Effect Transistors." *2026 Penn State Research Experience for Undergraduates (REU) Summer Symposium*. July 30, 2026
- **Stace, E.**, Smith, A., Lorinc, K. "Leveling Up: Resonant Scattering within Multi-Level Atoms." *2025 Summer Platform for Undergraduate Research (SPUR) Symposium*. July 31, 2025
- **Stace, E.**, Xu, S. "Numerical Measurement of Mean Free Path of High Energy Weakly Plasma." *2025 Annual Physics Undergraduate Research Poster Session*. April 25, 2025

## **Awards**

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- *Hilton Head Workshop 2026 Award for Conference Paper*: For top 4 papers at the conference
- *CLAS University Scholars Program*: Funded by University of Florida for undergraduate research during 2025-2026 academic year
- *DAP Travel Grant*: Travel reimbursement towards 2026 APS Global Physics Summit
- *Garrett Endowment for Physics Education* (4x)
- *2025 Erik Jonsson School of Engineering and Computer Science Award*: for exceptional poster at SPUR Symposium
- *2023 SIMIODE Challenge Using Differential Equations Modeling (SCUDEM) Outstanding Team Award*
- *Florida Bright Futures Scholarship*: 100% tuition coverage for four years

## **Technical Skills**

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**Programming Languages:** Python, C++, Verilog, Bash, HTML, CSS

**Libraries:** NumPy, Matplotlib, SciPy, H5Py, Pandas, Tensorflow, Sympy, Sunny.jl

**Laboratory Techniques/Equipment:** Network Analyzer, Wire Bonding, 4/2 -Point Probe Station, Mechanical Exfoliation, Scanning Electron Microscopy, Raman Spectroscopy, X-Ray Diffraction, Chemical Cleaning

**Cleanroom Techniques/Equipment:** Spin Benches, Liftoff Bath, Rapid Thermal Annealing, Photolithography

**Other Technologies:** KiCad, MATLAB, Vivado, Vitis, Quartus, Linux, Git, Github, SLURM, Jade,  $\LaTeX$